

PROJECT TITLE

AI-Based Breast Cancer Detection and Tissue Analysis Using Deep Learning

1. Problem Statement

Breast cancer is one of the major cancer burdens in India. Early and accurate identification of cancerous tissue can support timely diagnosis. However, examining large numbers of histopathology images manually can be time-consuming and depends heavily on expert assessment.

Problem: How can we use medical image analysis and deep learning to automatically identify cancerous breast tissue from histopathology images and provide reliable, data-driven insights?

2. Proposed Solution

We propose a deep-learning-based image classification system that analyzes breast tissue image patches and classifies them as:

- Healthy / Non-cancerous
- Cancerous / IDC-positive

The system will:

- Load and inspect the breast tissue image dataset.
- Explore healthy and cancerous tissue patches.
- Visualize tissue images and target distributions.
- Create PyTorch datasets and dataloaders.
- Train a CNN/deep-learning model.
- Validate the model using an appropriate validation strategy.
- Optimize the learning rate and training process.
- Analyze prediction errors.
- Generate probability maps showing areas where the model detects possible IDC.

3. Technology Stack

Area	Technology
Programming	Python
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Image Processing	PIL / OpenCV
Deep Learning	PyTorch
Model	CNN / Transfer Learning

4. Innovation

The innovation is not simply training a classifier. Your project can combine:

Image classification + tissue visualization + probability mapping + error analysis

The probability landscape of IDC is particularly useful because instead of only saying “cancer” or “no cancer,” the system can visualize regions where the model assigns a higher probability of cancerous tissue.

5. Data Analysis Component

Your notebook structure can become your complete analytical pipeline:

Dataset → EDA → Tissue Visualization → Target Analysis → Preprocessing → PyTorch Dataset → Model → Training → Validation → Evaluation → Error Analysis → IDC Probability Map

For example:

Exploratory Analysis

- Number of tissue images
- Number of healthy patches
- Number of IDC-positive patches
- Class imbalance
- Image dimensions
- Distribution of targets

Model Evaluation

- Confusion matrix
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- False positives and false negatives

6. Expected Impact

The project could help demonstrate how AI-assisted image analysis can:

- Reduce the amount of repetitive manual screening.
- Highlight potentially cancerous tissue regions for further examination.
- Provide quantitative analysis of histopathology images.
- Support research into automated breast-cancer detection.
- Make large-scale tissue-image analysis faster.